

Understanding the Effects of Short-Form Videos on Sustained Attention

Bei-Hong Lin*
neckson817@gmail.com
National Chung Cheng University
Chiayi County, Taiwan

Yu-Jung Chung*
eavaian0623@gmail.com
National Chung Cheng University
Chiayi County, Taiwan

Hao-Yuan Cheng
harrycheng1122@gmail.com
National Chung Cheng University
Chiayi County, Taiwan

Yu-Ting Yen
graislife8116@gmail.com
National Chung Cheng University
Chiayi County, Taiwan

Ching-Chuan Li
miller0809@gmail.com
National Chung Cheng University
Chiayi County, Taiwan

Fu-Yin Cherng
fuyincherng@gmail.com
National Chung Cheng University
Chiayi County, Taiwan

ABSTRACT

Understanding the rapidly expanding influences of short-form videos on cognition is critical. Previous studies have investigated the relationship between short-form video and several cognitive functions. However, few empirical studies have examined the effects of short-form video consumption on sustained attention. This study aimed to address these impacts through an online survey and a long-term field experiment. The online survey utilized a psychological test and found that an increased amount of short-form video consumption relates to poor sustained attention. The long-term field experiment clarified the directionality of the effect through long-term observation and sustained attention tests conducted at fixed intervals. We found that changing the duration of short-form video consumption does not significantly influence performance on most sustained attention tests. Our research constitutes a preliminary exploration that revealed the effects of short-form videos on sustained attention.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Short-form Video; Video Consumption; Sustained Attention; Long-term Field Experiment

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*The first and the second author contributed equally to this paper.

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1 INTRODUCTION

Short-form video applications are growing in popularity among people of all ages [57, 58]. These applications allow users to watch an endless stream of short-form videos (i.e., most are shorter than 2 minutes) [9, 25]. Moreover, short-form video applications provide brief, highly immersive content that suits the fragmented time available in modern life [15, 61]. Accordingly, short-form videos have gained popularity in recent years, with a growing number of users. For example, TikTok, the most popular short-form video application, had over 1 billion users in 2021, and its user base continues to expand [21].

With the increasing popularity of short-form videos, researchers have begun to understand how they influence behavior, emotions, and social interactions [28, 59, 65]. For example, prior studies found that short-form videos can fulfill various mental and social demands (e.g., entertainment, information needs, social relationships) [25, 58, 62]. However, other studies have indicated that short-form videos also lead to adverse effects, including addiction and decreased well-being [37, 54, 59, 63]. Some studies have also linked short-form video addiction to anxiety, stress, depression, and physical problems [28, 64].

Furthermore, recent research has begun to explore the impact of short-form videos on cognitive functions [7, 9]. Some studies have analyzed the design and core mechanism of short-form video applications and discovered that users tend to switch contexts more frequently than when watching longer-form videos due to the constant swiping involved in short-form video viewing. Such frequent context switching has been found to negatively impact individuals' cognitive abilities [31, 34, 56], particularly attention and memory [4, 27, 31]. For example, it has been observed that short-form videos and frequent context switching have a detrimental impact on memory [9]. Additionally, short-form video addiction is also related to attention deficit [7]. Due to this relation between short-form videos and attention, there is growing concern from the public that short-form videos could exacerbate negative effects on attention [53, 60].

However, attention, as defined by psychological theories [11], is multifaceted, comprising various subcomponents. Most recent research about the effect of short-form video on attention has treated attention as a monolithic construct rather than examining its distinct aspects [7, 18]. This oversimplified view may render it difficult to fully explain and interpret the nuanced phenomena associated

with short-form videos which are multidimensional in nature [24]. Consequently, misconceptions and biases in the research regarding the impacts of short-form video content on attention are possible.

In the hierarchical attentional model, sustained attention refers to the ability to maintain attentional focus on relevant stimuli that present repeatedly over extended periods [14, 26, 43], which is a common and essential form of attention in daily life [22, 47, 52]. Previous research indicates that frequent context switching may affect a person's sustained attention [1, 38, 56]. However, few studies have examined the link between short-form video and sustained attention. Furthermore, prior research only investigated the effect of short-form video in a laboratory and well-controlled setting [7, 9, 50]. Therefore, we aimed to investigate how watching short-form videos affects this cognitive ability in a daily and more realistic setting.

To explore this impact, we focused on daily watching habits associated with short-form videos and their relationship with sustained attention performance. Our first research question (RQ) is as follows: **RQ1: What is the correlation between the duration of daily short-form video consumption and sustained attention performance?** Moreover, we aimed to examine whether changes in the amount of short-form video consumption affect individuals' sustained attention performance. Hence, we investigated how short-form video consumption affects sustained attention by manipulating the duration of individuals' daily consumption of short-form video. The second research question is as follows: **RQ2: How does altering the duration of daily short-form video consumption impact sustained attention?** Finally, to determine whether the effects derive from the attributes of short-form video (i.e., frequent context switching) rather than video consumption in general, we compared the effects of short-form (video under 2 minutes [9, 25]) and long-form (video over 2 minutes) video consumption on sustained attention by increasing or decreasing participants' daily consumption. Our last research question, therefore, is the following: **RQ3: How do different types of videos (short-form versus long-form) impact sustained attention?**

To answer these research questions, we designed and conducted two studies. The first was an online survey with 84 participants; its aim was to explore the preliminary association between short-form video consumption and sustained attention. The second was a longitudinal field experiment (18 days) with 41 participants, in which we manipulated participants' duration of watching short-form and long-form videos to investigate the influence of habitual short-form video consumption on sustained attention.

Based on a joint analysis of the results from both the online survey and field experiment, our preliminary findings suggest there is a positive correlation between short-form video consumption and sustained attention deficit. However, it remains to be determined whether increased consumption of short-form videos can significantly impair an individual's sustained attention. Further investigation is required in the future to better understand this relationship. These combined studies address the knowledge gaps regarding short-form videos and long-term cognition in the current literature. Specifically, through a large-scale survey and longitudinal investigation, we gathered empirical evidence on how consuming more short-form videos affects individuals' sustained attention performance.

The results of the present study provide a foundation for future research to explore the influence of short-form video consumption on other types of attention and highlight the need to consider the potential effects of short-form videos on users' sustained attention when designing future applications.

2 ONLINE SURVEY

2.1 Data Collection

We conducted a web-based online survey to assess the correlation between people's daily consumption of short-form videos and their sustained attention performance. We recruited 84 participants on social media and online forums, consisting of 48 males and 36 females ($M = 31.33$, $SD = 13.67$). They watched an average of 31.5 minutes of short-form video ($SD = 36.1$), and most of the participants watched less than 60 minutes. We used the self-reported short-form video consumption and behavior task of sustained attention, namely the sustained attention to response task (SART) [6, 43, 48]. The SART is a well-studied psychological *go/no-go* task developed to evaluate an individual's sustained attention capacity.

As shown in Figure 1 (a), in each SART trial, participants were presented with a single digit (1-9), and they saw the digit appear alone in the center of the screen. The digits were categorized into two types of trials. For the *go* trials (i.e., digits 1, 2, and 4-9), participants were required to respond by pressing the space bar on the keyboard with their preferred hand. Conversely, participants were required to inhibit their impulse to respond during the *no-go* trials (i.e., digit 3). Based on calculations from prior related works [6, 48], *no-go* trials were relatively rare, constituting 11% of all trials. Each trial displayed a digit in a random font size (48, 72, 94, 100, or 120 pixels) for 250 ms. Then, the digit changed to a $\otimes$ symbol for 900 ms before the next trial begins.

To measure participants' sustained attention performance through the SART, we calculated their commission error score and reaction time variance (RTV) [2, 20, 33]. A commission error occurs when participants press the space key during a *no-go* trial. The higher the number of commission errors, the poorer the participant's sustained attention [2, 33]. In addition, we assessed inconsistency in an individual's response speed by calculating RTV. Previous literature suggests that a higher RTV represents higher distraction [20, 35]. The participants were rejected if they missed more than 10% of the *go* trials (21 trials) [6, 39, 48]. We collected 85 participants' data and rejected data from one of them.

2.2 Study Procedure

The online survey comprised two parts: the personal information questionnaire and the SART test. In the first part, participants were asked to complete a questionnaire regarding their daily consumption of short-form videos (e.g., perceived length of time spent watching short-form videos and the platforms they used). To ensure that the self-report duration was as precise as possible, we first asked participants to choose an interval zone from the five options (i.e., under 30 minutes, 31 to 60 minutes, 60 to 90 minutes, 91 to 120 minutes, or over 120 minutes). Next, they were asked to input the precise duration based on their memory. By comparing the responses to these two questions, it was possible to determine whether participants provided consistent answers about

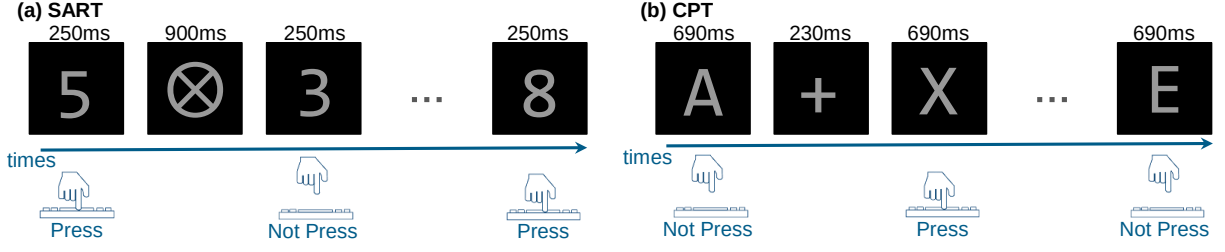


Figure 1: The process of the sustained attention to response task (SART) and the continuous performance test (CPT). (a) shows the SART process; in each trial of SART, the participant sees a single digit (1-9) for 250 ms. Then, the digit changes to a $\otimes$ symbol for 900 ms before the next trial begins. Participants must press the space bar on go trials (i.e., digits 1, 2, and 4-9) and refrain from pressing it on no-go trials (i.e., digit 3). (b) shows the CPT process; in each trial of CPT, the participant sees a single uppercase letter for 690 ms. Then, the digit changes to a “+” symbol for 230 ms before the next trial begins. Participants must press the space bar on go trials (i.e., X) and refrain from pressing it on no-go trials (i.e., A-W, Y, and Z).

their length of watching time. The complete survey and items can be found in Table 1 of the supplementary material. In the second part of the survey, participants took the SART, which measured their sustained attention capacity [6, 43, 48]. Participants were instructed to respond as quickly and accurately as possible to prevent them from changing their strategy to attain a higher score. Before they began the 225 formal experimental trials, they read the rules carefully and were required to complete 10 practice trials (including go and no-go trials) with no errors to ensure that they understood the SART.

2.3 Results and Discussion

We employed correlation analysis of the Pearson correlation coefficient to explore the relationship between short-form video consumption and sustained attention performance (i.e., commission error and RTV). Our results suggest a significantly positive correlation between the amount of short-form video consumption and the number of commission errors ($r(82) = .26, p < .05$). In addition, a significantly positive correlation was found between the duration of short-form video consumption and participants’ RTV ($r(82) = .25, p < .05$). The results reveal a significant correlation between short-form video consumption and SART performance. Since a greater number of commission errors are associated with difficulty inhibiting the impulse to respond and higher RTV indicates the inability to provide stable responses [2, 20, 33], it is likely that more time spent consuming short-form videos correlates to poorer sustained attention.

This relationship suggests that frequent consumption of short-form videos might potentially affect people’s daily sustained attention. These findings provide a preliminary insight into the relationship between short-form video consumption and sustained attention; however, it is important to note that correlation does not imply causation, which means the short-form video might not influence sustained attention. The other possibility is that people with worse sustained attention performance tend to watch more short-form videos. Consequently, it is necessary to explore the directionality of short-form video and sustained attention. To determine a cause-and-effect dynamic between short-form video consumption and sustained attention, we conducted a long-term field study by

manipulating the duration of daily short-form and long-form video consumption.

3 LONG-TERM FIELD EXPERIMENT

3.1 Experimental Design and Participants

This study explores how different amounts of short-form video consumption affect sustained attention. Moreover, it aims to determine how short-form and long-form videos affect sustained attention. As sustained attention improvement or reduction is a gradual cognitive process [17, 36], long-term observation is required to accurately detect any changes in sustained attention. Therefore, we carried out an 18-day field experiment involving randomized controlled field trials and online interviews to observe the gradual effects of short-form videos on sustained attention performance in daily life.

The field experiment employed a 2x2 between-subjects factorial design involving video types (short-form and long-form videos) and changes in video consumption (increasing and decreasing). The four conditions were labeled as short-form video increasing (SI), short-form video decreasing (SD), long-form video increasing (LI), and long-form video decreasing (LD). We recorded participants’ short-form and long-form video-watching time during the first three days and used them as their baselines for the short-form and long-form video-watching time in each condition (as shown in Figure 2). Based on their experimental condition, they increased or decreased their video-watching time by at least 30 minutes between the fourth day and the eighteenth day. For instance, participants in the SI condition were asked to increase their baseline time spent watching short-form videos by 30 minutes while maintaining their baseline time spent watching long-form videos.

To control the amount of change in video consumption in each condition, we chose a fixed amount as an experimental manipulation rather than a proportion based on each participant’s baseline for the following reasons. First, we wanted the amount of change to reach approximately 50% so that participants could significantly perceive the change in watching time in their daily routines. Based on the results of an online survey and previous research [30], it was found that most individuals watch short-form videos for less than 60 minutes per day. To ensure that most participants perceived

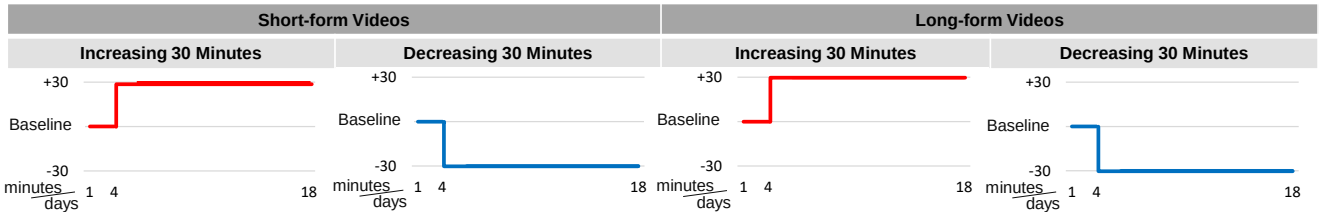


Figure 2: Experiment design of the long-term field experiment. The experiment had four conditions: watching short-form or long-form videos and increasing or decreasing viewing duration by 30 minutes. Participants’ average video-watching time from the first three days was the baseline. They increased or decreased their video-watching time by at least 30 minutes from day 4 to day 18.

the 50% change in video consumption, we therefore decided to establish half of 60 minutes as the fixed amount of change (i.e., 30 minutes). This fixed amount was established to prevent those who watched videos for more than 60 minutes per day at baseline from excessively increasing or decreasing their watching time.

We recruited 41 participants for the field experiment through public recruitment posts on social media and university forums. All participants were above 20 years old and had normal vision. The participants were randomly assigned to four conditions, resulting in 9 participants in condition SI (4 males; mean age = 24), 9 participants in condition SD (5 males; mean age = 23.2), 12 participants in condition LI (7 males; mean age = 24.8), and 11 participants in condition LD (7 males; mean age = 21.6). The detailed demographic is provided in supplementary material in Table 2.

3.2 Measurement

3.2.1 Behavioral Task Performance. In addition to the SART, we included the continuous performance task (CPT) to better assess the impact of short-form videos on task-based sustained attention [46]. The CPT is a well-studied tool that evaluates vigilance [11, 13, 26, 42], a crucial component of sustained attention that involves detecting and responding to infrequent stimuli [52]. As shown in Figure 1 (b), each CPT trial presents the participant with a single uppercase letter (A-Z) in the center of the screen. Although the CPT resembles the SART, participants must only press the space bar when the letter “X” appears for *go* trials, and for the other letters (i.e., A-W, Y, and Z) they are instructed to do nothing for the *no-go* trials. According to prior works [11, 13, 26, 42], *Go* trials must appear 8 times in every 31 trials. Each trial displays a letter for 690ms, then changes to a “+” symbol for 230 ms. As *go* trials are relatively rare in CPT, participants should only respond when a *go* trial appears and should not respond many times in the *no-go* trials. Participants were asked to complete 10 practice trials without errors before beginning the 310 formal experimental trials. We analyzed the standard deviation of reaction times (RTSD) and omission errors used to measure sustained attention over time [41].

3.2.2 Self-reported Questionnaire. We used two self-report questionnaires to determine participants’ perceived sustained attention performance, the attention-related cognitive errors scale (ARCES) [8] and the mindful attention awareness scale-lapse only (MAAS-LO) [5], both of which are intended to measure an individual’s

sustained attention [29, 32, 44, 51]. ARCES is a 12-item questionnaire that measures sustained attention failure in daily life (see Table 3 in the supplementary). The responses used a five-point Likert scale, and a higher score indicates a lower degree of sustained attention [8]. MAAS-LO, like ARCES, is a 12-item questionnaire that measures sustained attention lapses in daily life (see Table 4 in the supplementary). The responses used a six-point Likert scale, and a higher score also indicates poorer sustained attention [5].

3.2.3 Semi-structured Interview. We conducted the semi-structured interview to illuminate the physiological and psychological process of sustained attention change [12] (see Table 5 in the supplementary for the complete questions).

3.3 Procedure

3.3.1 Onboarding. After participants registered for the study, they were asked to digitally sign a consent form informing them about the experiment. It described the collected data and how it would be used. Next, they were instructed to watch an experiment video containing essential instructions for completing the daily report form and detailing the online test schedule. Then, they added a friend with the Line official experiment account for the following experiment.

3.3.2 Baseline Session. Our experiment consisted of the baseline and the manipulation sessions as shown in Figure 3. During the baseline (i.e., T1 to T3 in Figure 3), participants were asked to track their watching time for short-form and long-form videos over three days. They recorded the exact times they watched, the duration of watching time, and the applications they used for watching. In addition, they provided screenshots as evidence of their recordings. To ensure the accuracy of their reports, the experimenters cross-referenced the reported durations with the screenshots provided. If discrepancies were found, the participants were asked to adjust and resubmit their entries to ensure their watching time accurately reflected the evidence they provided.

3.3.3 Manipulation Session. After the participants submitted their record of the watching time on the third day, they were required to validate their baseline watching time. Participants were allowed to adjust their baseline if any special circumstances occurred during the baseline session. Once participants had confirmed their baseline watching times, we notified them of their experiment conditions. All participants were asked to modify their watching time

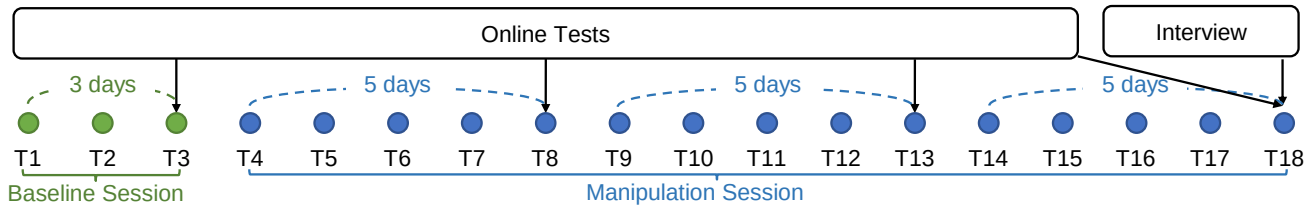


Figure 3: The long-term field experiment included two sessions: baseline and manipulation. Participants were in the baseline from T1 to T3. They then engaged in the manipulation session until T18. They were asked to track their video viewing time daily and complete the online tests every five days (T3, T8, T13, and T18). A semi-structured interview was conducted on T18.

for short-form or long-form videos based on the assigned condition. Throughout the manipulation session (i.e., T4 to T18 in Figure 3), participants were required to complete a daily report form, like the one used in the baseline.

3.3.4 Online Test and Interview. On the third, eighth, thirteenth, and eighteenth days (i.e., T3, T8, T13, and T18 in Figure 3), participants were instructed to complete the online test through Google Meet. The online test required them to complete two behavioral tasks (the SART and the CPT) and two self-reported questionnaires (ARCES and MAAS-LO); the total time of the online test lasted roughly 30 minutes. Participants were asked to share their screens during the SART and the CPT, ensuring they completed the tests correctly. On the eighteenth day, participants were required to complete an additional 10-minute semi-structured interview.

3.3.5 Ethical Considerations. To ensure all participants followed the instructions, we implemented a reward-and-punishment system. If they failed to reach the experiment goal or their watching time did not match the screenshots on a single occasion, they would not receive the additional reward. If they failed to take part in an online test or failed to follow the experiment conditions twice, they were excluded from the experiment and received \$8.05 USD per hour as compensation. Upon completing the field experiment, each participant was compensated \$32.29 USD for participation and an additional \$6.46 USD as a reward for following all the rules. Our institution review board (IRB) approved our field experiment procedure and design, and all participants were asked to review and agree to the IRB consent form.

3.4 Results and Discussion

To evaluate the effect of video consumption on sustained attention over time, we analyzed the differences in measurements taken at T8, T13, and T18, comparing them with the baseline established at T3, as shown in Figure 4. To examine whether the consumption of short-form videos is associated with sustained attention, we employed a one-sample t-test to separately compare these measurements in the manipulation session (i.e., T8, T13, T18) with those from the baseline session (i.e., T3) in the short-form video conditions. The detailed statistical results are presented in Table 6 and Table 7 of the supplementary material.

As shown in Figure 4 (a), our results show that the differences in SART commission errors were significantly lower than zero for both SD ($t(26) = -3.670, p < .001$) and SI ($t(26) = -4.852, p < .001$).

The results demonstrate that the difference between the number of SART commission errors in the manipulation and those in the baseline session is negative, which means that participants in the SI and SD conditions tended to diminish their errors on the SART. Similarly, Figure 4 (d) indicates that the difference in CPT RTSD was also significantly lower than zero in both SI and SD conditions ($t(26) = -2.417, p < .05$; $t(26) = -2.757, p < .01$). These results indicated that during the manipulation session, participants' performance on the SART and CPT improved compared to the baseline session, regardless of whether the duration of short-form video consumption increased or decreased.

However, we did not observe the same improvements regarding participants' perceived sustained attention (as measured by ARCES and MAAS-LO). This discrepancy may be due to a learning effect inherent in task-based sustained attention, which is not present in self-report questionnaires. However, the SART and CPT have been proven stable over time [43], making them reliable for repetitive testing in various studies [23, 45]. Further research is needed to understand whether the learning effects exist and the underlying reasons.

To examine the impacts of different changes in video consumption (i.e., increasing or decreasing) and video types (i.e., short-form and long-form video) on sustained attention, we used a non-parametric two-way ANOVA on the SART RTV and CPT omission error, as they failed to pass the normality tests (Shapiro-Wilk test: $p < .001$) [10]. We performed two-way ANOVA on the rest of the measurements, which passed the normality tests. The detailed statistical results are presented in Table 8 and Table 9 of the supplementary material.

For the effect of changes in video consumption, our results revealed that increasing video consumption has marginally significantly higher ARCES scores than decreasing video consumption. ($F(1, 119) = 3.158, p = .078$). These results demonstrated that increasing the duration of video consumption tended to degrade sustained attention, while reduced consumption improved it. However, non-significant correlations were found for MAAS-LO. Therefore, based on self-reported questionnaires, we cannot infer a definite relationship between video consumption and sustained attention.

As for the influence of video types, we found a significant effect of video type on CPT omission errors ($F(1, 118) = 3.930, p < .05$) and a marginally significant effect on SART RTV ($F(1, 118) = 3.206, p = .075$). However, further exploration is needed to understand the underlying mechanisms. Lastly, we found no significant interaction

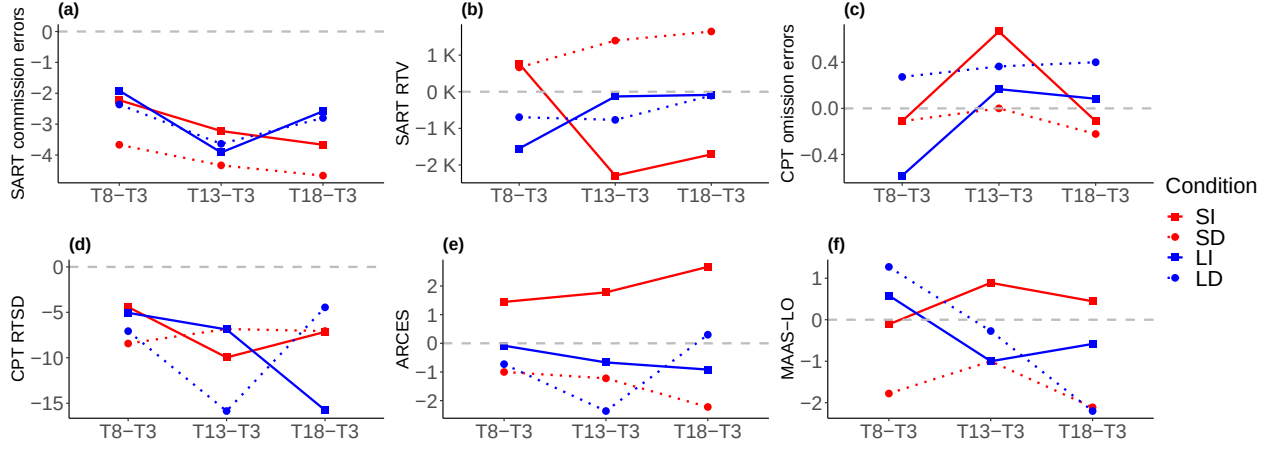


Figure 4: Results of the long-term field experiment. In each line graph, the x-axis indicates the three time points, including T8-T3, T13-T3, and T18-T3. The y-axis indicates the difference between the value in the manipulation session (T8, T13, T18) and the baseline session (T3). A horizontal dotted line depicts the zero point of the difference. (a) shows SART commission errors; (b) shows SART RTV; (c) shows CPT omission errors; (d) shows CPT RTSD; (e) shows ARCES; and (f) shows MAAS-LO.

between the two factors in both perceived sustained attention and task-based performance.

4 GENERAL DISCUSSION AND FUTURE WORKS

For the online survey, we found a significant negative correlation between short-form video consumption and the behavioral task of sustained attention. The results match the previous study, which focused on the relationship between short-form video consumption and self-reported questionnaires of sustained attention [50]. Based on our task-based findings and self-reported results from prior research, we infer that viewing short-form videos is related to the ability to maintain attentional focus. In addition, the previous studies demonstrated that excessively consuming short-form video is related to poor attention [7], which supports our inference above. However, although the association between short-form video consumption and sustained attention is evident, it is unclear whether watching them leads to longitudinal sustained attention inability in daily life, which is limited to be observed in short-period study and laboratory environments [7, 9, 50].

Hence, we conducted a long-term field study to clarify the directionality of short-form video consumption and sustained attention. We found that increasing the consumption duration of either short-form or long-form videos may lead to poorer sustained attention. However, the effects could only be observed in the self-report metric (ARCES). The findings resemble those of previous studies concerning frequent context switching in media use. While frequent context switching was found to have a more significant correlation in self-reported sustained attention, it yielded mixed results in task-based sustained attention [31, 38, 39, 55]. Based on these findings, we conclude that the negative impact of increased video consumption may be reflected only in individuals' subjective perceptions of sustained attention rather than their task-based sustained attention.

In addition, previous studies demonstrated that problematic use of short-form videos, such as addiction, leads to attentional deficit instead of watching more duration of short-form video consumption [3, 49]. This implies that video consumption alone may not be detrimental to an individual's sustained attention unless it is excessive or compulsive. In conclusion, we found that watching short-form videos was associated with negative effects on sustained attention, and increasing video consumption might slightly degrade perceived sustained attention. Current preliminary findings suggest that individuals with worse sustained attention might tend to watch more short-form videos. However, more studies are needed to verify the above conclusion.

Our study acknowledges several limitations. First, we conducted an online field study to investigate sustained attention. Although it can improve the generalization, it might lack internal validity because confounding variables are not fully controlled [16, 40]. Future studies should replicate our research in a more controlled setting to ensure the effects of short-form video on sustained attention. Secondly, our 18-day field experiment might have been too short to accurately measure the effects of watching short-form videos on sustained attention. Future researchers should consider extending the duration of their experiments.

In addition, although quantitative statistics can reveal the effects of video consumption on sustained attention, it may be difficult to interpret the underlying reasons for these effects in our field experiment. Hence, we plan to analyze the qualitative results of the semi-structured interview to explore the underlying reasons for the sustained attention change. Next, as there are not enough participants in each condition of the field experiment, we plan to recruit additional participants to increase the statistical power for future work. Finally, we consider that the experience sampling method (ESM) [19] would be an effective research approach to gain deeper insights into the effects of short-form videos on sustained attention. Future research can develop mobile applications with combinations

of different interactive designs from short-form video platforms. By collecting data on users' behavioral logs and reports on sustained attention while using these applications, future researchers can explore how various interactive designs influence sustained attention behaviorally and perceptually. This data can also provide design implications for researchers and designers to minimize the negative impacts of short-form videos on sustained attention, potentially enhancing user experience with such content.

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